

Geometric Sequences

Algebra Worksheet · Grade 9–11

Name: _____

Date: _____

Learning Objectives

- Identify whether a sequence is geometric by finding a common ratio
- Apply the geometric sequence formula to find specific terms
- Solve for unknown values such as the first term, common ratio, or term number

Problems

1. Find the common ratio of the geometric sequence below.

2, 6, 18, 54

2. Determine whether the following sequence is a geometric sequence. Explain why or why not.

4, 8, 16, 32

3. Determine whether the following sequence is a geometric sequence. Explain why or why not.

3, 6, 10, 15

4. Find the 6th term of the geometric sequence below using the formula $a_n = a_1 \times r^{n-1}$.

2, 6, 18, 54, ...

5. Find the 8th term of the geometric sequence below.

5, 10, 20, 40, ...

6. Find the 5th term of the geometric sequence with a first term of 64 and a common ratio of one half.

$a_1 = 64$. $r = \frac{1}{2}$. $n = 5$



7. A geometric sequence has a first term of 7 and a common ratio of 3. Find the 7th term.

$$a_1 = 7, r = 3, n = 7$$

8. The 1st term of a geometric sequence is 4 and the 4th term is 108. Find the common ratio.

$$a_1 = 4, a_4 = 108$$

9. Which term of the geometric sequence below equals 1458?

2, 6, 18, 54, ...

10. A ball is dropped from a height of 81 meters. Each bounce reaches two thirds of the previous height. Find the height of the 5th bounce.

$$a_1 = 81, r = \frac{2}{3}, n = 5$$



Geometric Sequences — Answer Key

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Answer Key

1. Answer: $r = 3$

- Divide the second term by the first: $6 \div 2 = 3$
 - Verify: $18 \div 6 = 3$ and $54 \div 18 = 3$
 - The common ratio is $r = 3$
-

2. Answer: Yes; $r = 2$

- Divide consecutive terms: $8 \div 4 = 2$
 - $16 \div 8 = 2$ and $32 \div 16 = 2$
 - All quotients are equal, so the sequence is geometric with $r = 2$
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3. Answer: No; no common ratio

- Divide consecutive terms: $6 \div 3 = 2$
 - $10 \div 6 \approx 1.67$ and $15 \div 10 = 1.5$
 - The ratios are not equal, so this is NOT a geometric sequence
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4. Answer: $a_n = 1458$

- Identify: $a = 2$, $r = 3$, $n = 6$
 - Apply formula: $a_n = 2 \times 3^{(6-1)} = 2 \times 3^5$
 - $3^5 = 243$
 - $a_n = 2 \times 243 = 1458$
-

5. Answer: $a_n = 640$

- Identify: $a = 5$, $r = 2$, $n = 8$
 - Apply formula: $a_n = 5 \times 2^{(8-1)} = 5 \times 2^7$
 - $2^7 = 128$
 - $a_n = 5 \times 128 = 640$
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6. Answer: $a_n = 4$

- Apply formula: $a_n = 64 \times (1/2)^{(5-1)} = 64 \times (1/2)^4$
 - $(1/2)^4 = 1/16$
 - $a_n = 64 \times (1/16) = 4$
-

7. Answer: $a_n = 5103$

- Apply formula: $a_n = 7 \times 3^{(7-1)} = 7 \times 3^6$
 - $3^6 = 729$
 - $a_n = 7 \times 729 = 5103$
-

8. Answer: $r = 3$

- Use formula: $a_n = a_1 \times r^{(n-1)}$





- $108 = 4 \times r^3$
- $r^3 = 108 \div 4 = 27$
- $r = \sqrt[3]{27} = 3$

9. Answer: $n = 7$ (the 7th term)

- Identify: $a = 2$, $r = 3$, $a_n = 1458$
- Set up equation: $1458 = 2 \times 3^{(n-1)}$
- Divide both sides by 2: $3^{(n-1)} = 729$
- Since $3^6 = 729$, then $n - 1 = 6$, so $n = 7$

10. Answer: $a_n = 16$ meters

- Apply formula: $a_n = 81 \times (2/3)^{(5-1)} = 81 \times (2/3)^4$
 - $(2/3)^4 = 16/81$
 - $a_n = 81 \times (16/81) = 16$
 - The height of the 5th bounce is 16 meters
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